

2026.4.7. ODE I.

1. 初等积分求解 ODE.

① $y' = \frac{x+y}{x-y}$.

② 这是关于 x, y 的齐次函数.

令 $z = \frac{y}{x}$. $y = xz$. $y' = z + xz'$.

原方程化为 $x \frac{dz}{dx} = \frac{z^2+1}{1-z}$

$$\Rightarrow \frac{1-z}{z^2+1} dz = \frac{dx}{x} \Rightarrow \arctan z = C + \ln |x| \cdot \sqrt{1+z^2} \\ = C + \ln \sqrt{x^2+y^2}$$

$$\Rightarrow \sqrt{x^2+y^2} = C e^{\arctan \frac{y}{x}} \quad (C > 0).$$

(法2.) $\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$

原式变成 $dr + r d\theta = 0$.

$$\Rightarrow r = C e^{\theta} \quad (C > 0)$$

$$\text{即 } \sqrt{x^2+y^2} = C e^{\arctan \frac{y}{x}} \quad (C > 0)$$

该轨迹称为螺旋线.

② $2x^3 y' = (2x^2 - y^2) y$. ($x > 0$).

$$y' = \frac{(2x^2 - y^2) y}{2x^3}, \quad \text{这是齐次函数.}$$

令 $u = \frac{y}{x}$. $y' = u + x u'$

$$\Rightarrow \cancel{2u + 2xu} \quad 2x^3 u + 2x^4 u' = (2 - u^2) u x^3$$

$$\Rightarrow 2x u' = -u^3 \Rightarrow \frac{dx}{x} = -\frac{2du}{u^3}$$

$$\Rightarrow C + \ln x = \frac{1}{u^2} \Rightarrow x = C e^{\frac{x^2}{y^2}} \quad (C > 0)$$

($x > 0$)

$$\textcircled{3} \quad y' = \frac{-x+2y+5}{2x-y-4}$$

这是形如 $y' = f\left(\frac{ax+by+c}{dx+ey+f}\right)$ 且 $\begin{vmatrix} a & b \\ d & e \end{vmatrix} \neq 0$.

~~令 $u = x$~~ 因此, 方程 $\begin{cases} -x+2y+5=0 \\ 2x-y-4=0 \end{cases}$ 有解 $\begin{cases} x=1 \\ y=-2 \end{cases}$ 平移后齐次.

即 $\begin{cases} -(x-1) + 2(y+2) = -x+2y+5 \\ 2(x-1) - (y+2) = 2x-y-4 \end{cases}$

故换元 $\begin{cases} u = x-1 \\ v = y+2 \end{cases} \quad y' = \frac{dy}{dx} = \frac{dv}{du}$

$\Rightarrow \frac{dv}{du} = \frac{2v-u}{2u-v}$ 是齐次函数.

令 $t = \frac{v}{u}$. $v = tu$. $\frac{dv}{du} = t + u \frac{dt}{du}$

$\Rightarrow t + u \frac{dt}{du} = \frac{2t-1}{2-t} \Rightarrow u \frac{dt}{du} = \frac{2t-1-2t+t^2}{2-t} = \frac{t^2-1}{2-t}$

$\Rightarrow \frac{(2-t)dt}{t^2-1} = \frac{du}{u}$ 两边积分. $\int \frac{du}{u} = \ln|u| + C$.

$$\begin{aligned} \int \frac{(2-t)dt}{t^2-1} &= \int \frac{2dt}{t^2-1} - \int \frac{tdt}{t^2-1} \\ &= \int \left(\frac{1}{t-1} - \frac{1}{t+1} \right) dt - \frac{1}{2} \int \frac{dt^2}{t^2-1} = \ln \left| \frac{t-1}{t+1} \right| - \frac{1}{2} \ln|t^2-1|. \\ &= \ln \sqrt{\left| \frac{t-1}{(t+1)^3} \right|}. \end{aligned}$$

因此 $\ln \sqrt{\left| \frac{t-1}{(t+1)^3} \right|} = \ln|u| + C$. 两边 $\times 2$ 再取指数
有绝对值打开, 有可能负.

$\Rightarrow \frac{t-1}{(t+1)^3} = Cu^2$. ($C \neq 0$). Recall. $u = x-1$

$\Rightarrow (x-1)^2 \cdot \frac{-x+y+3}{(x+y+1)^3} = C(x-1)^2$

$\Rightarrow -x+y+3 = C \cdot (x+y+1)^3$. ($C \neq 0$)

另检查 $C=0$ 和 $x+y+1=0$. 也是解 $\star$

全部解是 $-x+y+3 = C \cdot (x+y+1)^3$, ($C \in \mathbb{R}$) 和 $x+y+1=0$

$$\textcircled{4} \quad y' = \frac{x+2y+1}{2x+4y-1}.$$

这是形如 $y' = f\left(\frac{ax+by+c}{dx+ey+f}\right)$ 且 $\begin{vmatrix} a & b \\ d & e \end{vmatrix} = 0$ 情形。

无法平移变齐次，直接换元 $u = x+2y$ 。

$$\frac{du}{dx} = 1 + 2y' \Rightarrow y' = \frac{u'-1}{2}.$$

$$\Rightarrow u' - 1 = 2 \cdot \frac{u+1}{2u-1} = \frac{2u+2}{2u-1} = 1 + \frac{3}{2u-1}.$$

$$\Rightarrow u' = 1 + \frac{2u+2}{2u-1} = \frac{4u+1}{2u-1}.$$

$$\Rightarrow \frac{2u-1}{4u+1} du = dx.$$

$$\begin{aligned} \Rightarrow x + C &= \int \frac{2u-1}{4u+1} du = \int \frac{1}{2} + \frac{-\frac{3}{2}}{4u+1} du \\ &= \frac{u}{2} - \frac{3}{8} \ln|4u+1| \\ &= \frac{x}{2} + y - \frac{3}{8} \ln|4x+8y+1|. \end{aligned}$$

$$\Rightarrow \ln|4x+8y+1| = \frac{8}{3} \left(y - \frac{x}{2}\right) + C.$$

$$\Rightarrow 4x+8y+1 = C e^{\frac{8}{3}(y-\frac{x}{2})} \quad (C \neq 0)$$

检查 $4x+8y+1=0$ 也是解

$$\Rightarrow \text{全部解是 } (4x+8y+1)^3 = C e^{8y-4x} \quad (C \in \mathbb{R})$$

$$\textcircled{5} \quad y'(x+2y) = 1. \quad \text{求 } y_{(0)} = -1 \text{ 的特解.}$$

$$\text{即 } y' = \frac{1}{x+2y} \quad \text{形如 } y' = f\left(\frac{ax+by+c}{dx+ey+f}\right)$$

$$\text{换元 } u = x+2y. \quad u' = 1+2y'. \quad y' = \frac{u'-1}{2}$$

$$\Rightarrow u' - 1 = \frac{2}{u} \Rightarrow \frac{du}{dx} = 1 + \frac{2}{u} = \frac{u+2}{u}$$

$$\begin{aligned} \Rightarrow dx &= \frac{u du}{u+2} \Rightarrow \int dx = \int \frac{u du}{u+2} \\ x + C &= u - 2 \ln|u+2| \\ &= x+2y - \ln(x+2y+2)^2 \end{aligned}$$

$$\Rightarrow 2y = C + \ln(x+2y+2)^2. \quad (C \in \mathbb{R}).$$

代入 $(x, y) = (0, -1)$: $-2 = C + \ln 0^2$ 不在该通解内!

57 r-1

检查过程. 唯一不必要的步骤是除以 $u+2$.

因为有可能 $u+2=0$!

$$u+2=0 \text{ 对应 } x+2y+2=0. \text{ 即 } y=-1-\frac{x}{2}.$$

检验: $x=0$ 时, $y=-1$

$$y' = -\frac{1}{2}, \quad x+2y = -2 \Rightarrow y'(x+2y) = 1 \checkmark$$

因此所求特解是 $x+2y+2=0$.

Rmk. 这称为奇解/包络. 图像上是通解曲线族的一条包络直线.

$$\textcircled{6} \quad x^2 y' - \cos 2y = 1. \quad y(+\infty) = \frac{9}{4}\pi.$$

$$\text{解: } \frac{dx}{x^2} = \frac{dy}{1+\cos 2y} = \frac{1}{2} \sec^2 y dy$$

$$\Rightarrow -\frac{1}{x} = \frac{1}{2} \tan y + C. \quad \text{代入 } (x, y) = (+\infty, \frac{9}{4}\pi).$$

$$\text{得: } 0 = \frac{1}{2} \tan \frac{9}{4}\pi + C \\ = \frac{1}{2} + C \Rightarrow C = -\frac{1}{2}.$$

$$\Rightarrow \text{所求为 } \tan y - 1 + \frac{2}{x} = 0.$$

$$\text{即 } y = 2\pi + \arctan \frac{x-2}{x} \quad (x > 0).$$

Rmk. 所求通解是 $y = k\pi + \arctan \frac{Cx-2}{x} \quad (k \in \mathbb{Z}, C \in \mathbb{R}).$

这不违反一阶 ODE 通解仅含一个常数 C 的事实. 因为通积分是 $\tan y + \frac{2}{x} + C = 0 \quad (C \in \mathbb{R}).$ k 的引入只是将该曲线分成不同的连续部分.

另外, 从该结构可见. $y(+\infty) = k\pi + \arctan C$

$$y(-\infty) = k\pi - \arctan C.$$

这些曲线被限制在 $y = (k \pm \frac{1}{2})\pi$ 之间, 这导致

$y = (k + \frac{1}{2})\pi, k \in \mathbb{Z}$ 都是奇解. 对应 C 取 $+\infty$.

⑦ $3y^2y' + 16x = 2xy^3$. $x \rightarrow +\infty$ 时, y 有界.

$\frac{3y^2 dy}{y^3 - 8} = 2x dx$ 不充要 $\Rightarrow y=2$ 是一个解

$\Rightarrow \ln|y^3 - 8| = C + x^2 \Rightarrow y = (8 + Ce^{x^2})^{\frac{1}{3}} (C \neq 0)$

综合 $\Rightarrow y = (8 + Ce^{x^2})^{\frac{1}{3}} (C \in \mathbb{R})$ 包含了 $y=2$.

欲有界 $\Leftrightarrow C=0$. 故所求为 $y=2$.

⑧ (2024.9.)

$f: \mathbb{R} \rightarrow \mathbb{R}$ 可导. $f'(x) = xf(x) + x \int_0^1 t f(t) dt$.

记 $\int_0^1 t f(t) dt = C$. 即 $f' = x(f + C)$

$\Rightarrow \frac{df}{f+C} = x dx \Rightarrow \ln(f+C) = \frac{x^2}{2} + \tilde{C}$

$\Rightarrow f(x) = C_1 e^{\frac{x^2}{2}} - C_2$ 这里 $C_1 \neq 0$
 $C_2 = \int_0^1 t f(t) dt$
 会失去常值解.

(i) $\int_0^1 x f(x) dx = C_1 \int_0^1 x e^{\frac{x^2}{2}} dx - C_2 \int_0^1 x dx$
 $= C_1 (\sqrt{e} - 1) - \frac{1}{2} C_2 = C_2$

$\Rightarrow C_2 = \frac{2}{3} C_1 (\sqrt{e} - 1)$.

$\Rightarrow f(x) = C \cdot (e^{\frac{x^2}{2}} - \frac{2}{3} (\sqrt{e} - 1))$. ($C \neq 0$)

(ii) 检查常值解. $f(x) = -C = -\int_0^1 t f(t) dt$.

那么, $C = \int_0^1 t f(t) dt = -\int_0^1 C t dt = -\frac{1}{2} C \Rightarrow C = 0$

综上: $f(x) = C \cdot (e^{\frac{x^2}{2}} - \frac{2}{3} (\sqrt{e} - 1))$. ($C \in \mathbb{R}$)

2. 一元积分因子.

对形如 $f'(x) + g(x)f(x) = 0$ 型方程. 同乘 $h(x)$

$$\Rightarrow h(x)f'(x) + h(x)g(x)f(x) = 0.$$

使左侧变成一个乘积型导数. $(h(x)f(x))'$

$$\text{则 } (h(x)f(x))' = h(x)f'(x) + \underline{h'(x)f(x)}$$

$$= h(x)f'(x) + \underline{h(x)g(x)f(x)}$$

要求满足 $h'(x) = h(x)g(x)$ 即 $(\ln h(x))' = g(x)$

$$\text{故 } \ln h(x) = \int_a^x g(t) dt + C$$

$$\Rightarrow h(x) = C e^{\int_a^x g(t) dt} \quad \text{特别地, 取 } C=1 \text{ 即可.}$$

$$\text{Eg. } f'(x) + f(x) \geq 0, \quad f(0) \geq 0. \quad \text{则 } f(x) \geq 0, \quad \forall x \geq 0.$$

$$\text{证明: } (e^x f(x))' \geq 0 \Rightarrow f(x) \geq e^{-x} f(0) \geq 0. \quad \square$$

$$\text{Eg. } f'(x) + \frac{1}{x} f(x) = 0.$$

$$h(x) = e^{\int_1^x \frac{1}{t} dt} = e^{\ln x} = x.$$

$$\Rightarrow x f'(x) + f(x) = 0 \Rightarrow (x f(x))' = 0$$

$$\Rightarrow x f(x) = C \Rightarrow f(x) = \frac{C}{x}.$$

$$\text{Eg. } f'(x) + \frac{1}{1+x^2} f(x) = 0.$$

$$h(x) = e^{\int_0^x \frac{dt}{1+t^2}} = e^{\arctan x}$$

$$\Rightarrow (e^{\arctan x} f(x))' = 0 \Rightarrow f(x) = C e^{-\arctan x}$$

[Gronwall 不等式].

$$g(x) \geq 0. \quad f(x) \leq c + \int_a^x f(t)g(t)dt. \quad \text{则 } f(x) \leq ce^{\int_a^x g(t)dt}$$

证明: 令 $F(x) = \int_a^x f(t)g(t)dt$.

$$F'(x) = f(x)g(x) \leq (c + F(x))g(x) \quad F, F' \text{ 同时出现}$$

$$\Rightarrow \underline{F'(x) - g(x)F(x)} \leq cg(x) \quad \text{两边乘积分因子. 令 } e^{-\int_a^x g(t)dt}$$

$$\Rightarrow \left(F(x) \cdot e^{-\int_a^x g(t)dt} \right)' \leq cg(x) e^{-\int_a^x g(t)dt} \\ = -c \cdot \left(e^{-\int_a^x g(t)dt} \right)'$$

$[a, x]$ 上
积分: $F(x) \cdot e^{-\int_a^x g(t)dt} - \underbrace{F(a)}_{F(a)=0} \cdot e^0 \leq -c \cdot (e^{-\int_a^x g(t)dt} - e^0).$

$$\Rightarrow F(x) \leq e^{\int_a^x g(t)dt} \cdot c(1 - e^{-\int_a^x g(t)dt}) \\ = c \cdot (e^{\int_a^x g(t)dt} - 1).$$

$$\Rightarrow f(x) \leq c + F(x) \leq c \cdot e^{\int_a^x g(t)dt}. \quad \square.$$

应用举例. ① Picard 定理唯一-性部分.

如果 $\begin{cases} y' = f(x, y) \\ y(x_0) = y_0 \end{cases}$ 满足 $|f(x, y_1) - f(x, y_2)| \leq L \cdot |y_1 - y_2|$
则解是唯一的.

证明: 解满足 $y(x) = y_0 + \int_{x_0}^x f(t, y(t))dt$.

设有 2 个解 $y_1(x), y_2(x)$.

$$\text{则 } |y_1(x) - y_2(x)| = \left| \int_{x_0}^x f(t, y_1(t)) - f(t, y_2(t)) dt \right| \\ \leq \int_{x_0}^x |f(t, y_1(t)) - f(t, y_2(t))| dt \\ \leq L \cdot \int_{x_0}^x |y_1(t) - y_2(t)| dt.$$

这就是 Gronwall 不等式 $g(x) \equiv L \geq 0, c = 0$ 情形.

因此, $|y_1(x) - y_2(x)| \leq 0 \Rightarrow y_1(x) \equiv y_2(x). \quad \square.$

② 将一般 ODE 转化为恰当方程.

(i) $(x^2+y^2+x) dx + y dy = 0.$

同乘因子 $f(x)$. $f(x)(x^2+y^2+x) dx + f(x)y dy = 0$

欲使之为恰当的, 要求 $\frac{\partial}{\partial y}(f(x)(x^2+y^2+x)) = \frac{\partial}{\partial x}(f(x)y)$

即 $2yf(x) = f'(x)y$

即 $f'(x) - 2f(x) = 0$

$\Rightarrow f(x) = e^{2x}.$

(ii) $(3x^2y+2xy+y^3) dx + (x^2+y^2) dy = 0.$

同乘 $f(x)$. 令 $\frac{\partial}{\partial y}(f(x)(3x^2y+2xy+y^3)) = \frac{\partial}{\partial x}(f(x)(x^2+y^2))$

即 $(3x^2+2x+y^2)f(x) = 2xf(x) + x^2f'(x) + y^2f'(x)$

$\Leftrightarrow \cancel{f'(x) - 3(1+\frac{y^2}{x^2})f(x)}$

$f'(x) - 3f(x) = 0 \Rightarrow f(x) = e^{3x}.$

③ 积分因子的复杂选择.

$y^2(ydx-2xdy) = x^3(xdy-2ydx).$

同乘 $x^a y^b$ 待定系数.

$(x^a y^{b+3} + 2x^{a+3} y^{b+1}) dx - (2x^{a+1} y^{b+2} + x^{a+4} y^b) dy = 0.$

$\downarrow$
 $\frac{\partial}{\partial y}(\cdot) = (b+3)x^a y^{b+2} + 2(b+1)x^{a+3} y^b$

$\downarrow$
 $\frac{\partial}{\partial x}(\cdot) = -2(a+1)x^a y^{b+2} - (a+4)x^{a+3} y^b.$

令之相等: $\begin{cases} b+3 = -2(a+1) \\ 2(b+1) = -(a+4) \end{cases} \Leftrightarrow \begin{cases} a = -\frac{4}{3} \\ b = -\frac{7}{3} \end{cases}$

$\Rightarrow (x^{-\frac{4}{3}} y^{\frac{2}{3}} + 2x^{\frac{5}{3}} y^{-\frac{4}{3}}) dx - (2x^{-\frac{1}{3}} y^{-\frac{1}{3}} + x^{\frac{8}{3}} y^{-\frac{7}{3}}) dy = 0.$

积分. $-3x^{-\frac{1}{3}} y^{\frac{2}{3}} + \frac{3}{4} x^{\frac{8}{3}} y^{-\frac{4}{3}} = C.$